DO NOW — RECALL

Lesson 143 · The TCP/IP Stack and Encapsulation

Name: _____ Class: _____ Date: _____

1. Which device typically forwards traffic between different IP networks?

2. Which logic gate outputs 1 only when both inputs are 1?

3. What is the accumulator for?

4. Which is faster in the worst case on sorted arrays, linear or binary search?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Router.

2. Model answer

AND.

3. Model answer

Holding a CPU's working data/ALU results.

4. Model answer

Binary search: O(log n) rather than O(n).

By the end, you can…

• HTTP sends a request body/application message.

• TCP segments carry ports, IP packets carry network addresses.

• Ethernet/Wi-Fi frame delivers packet over one link, then changes at next hop.

Vocabulary — know these terms

application layer — Protocols providing services to programs, such as HTTP or DNS.

transport layer — End-to-end service such as TCP or UDP.

internet layer — IP packet addressing and routing between networks.

link layer — Local frame delivery on physical or wireless link.

NEW LEARNING

Read, annotate and explain

Application data

HTTP sends a request body/application message.

Transport and IP

TCP segments carry ports, IP packets carry network addresses.

Frame on local link

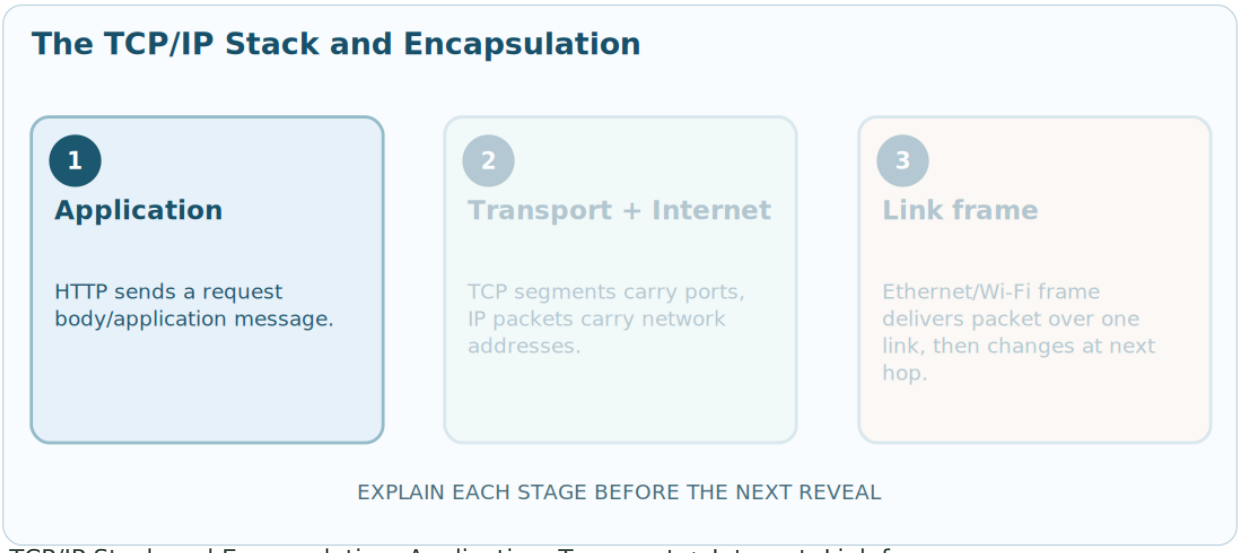
Ethernet/Wi-Fi frame delivers packet over one link, then changes at next hop.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

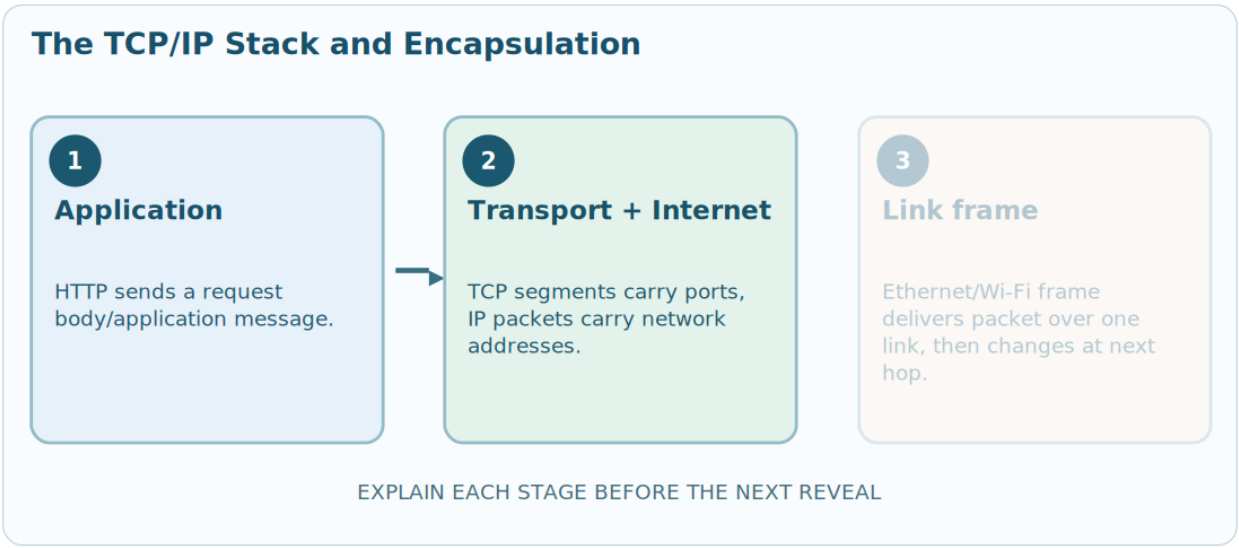
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



The TCP/IP Stack and Encapsulation: Application, Transport + Internet, Link frame

New Learning

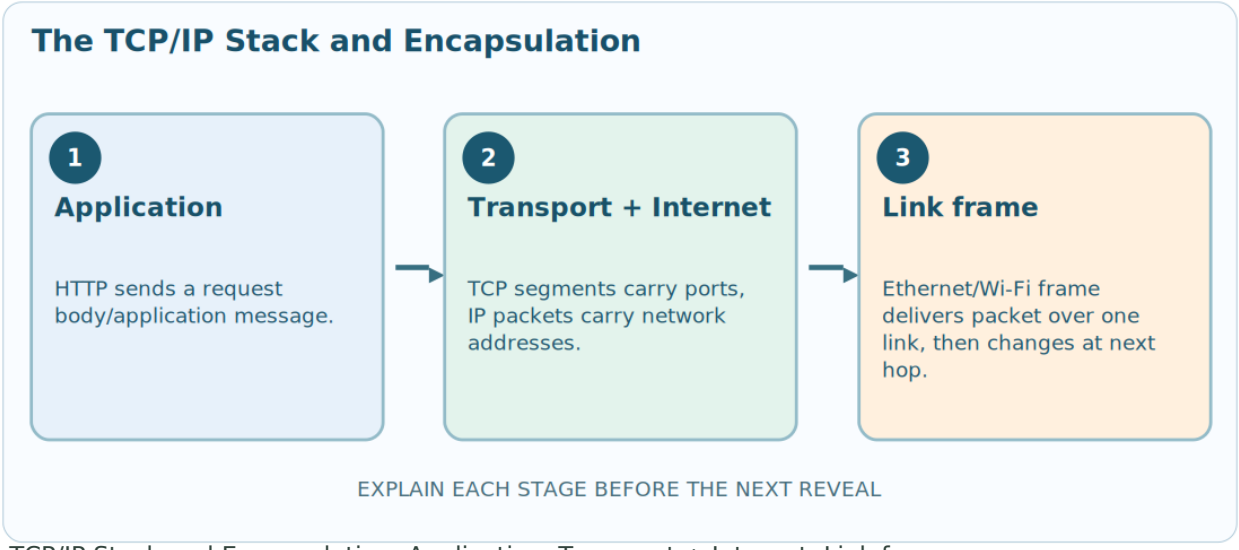


The TCP/IP Stack and Encapsulation: Application, Transport + Internet, Link frame

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



The TCP/IP Stack and Encapsulation: Application, Transport + Internet, Link frame

I DO — WORKED EXAMPLE

Follow the worked method

A browser sends an HTTP request using TCP/IP over Ethernet. Describe encapsulation from top to bottom.

HTTP application data

Identify the relevant information and make a prediction.

TCP segment in IP packet

Explain the change using correct technical terms.

Ethernet frame

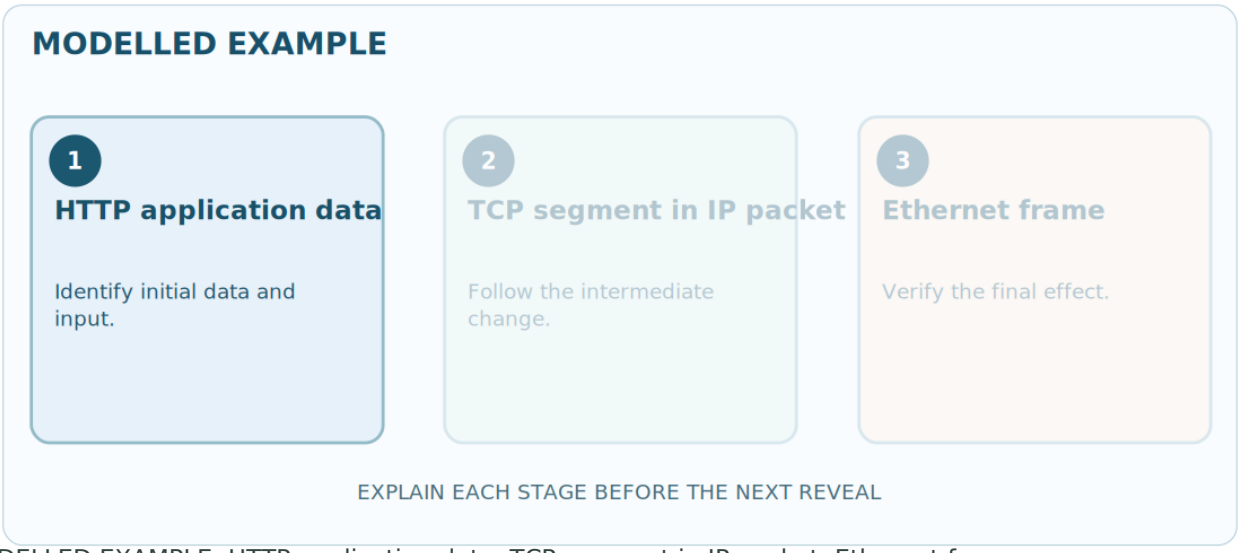
HTTP application bytes become TCP segment payload; the TCP segment goes into an IP packet; the packet is carried in an Ethernet frame. Receivers process headers in reverse to deliver data to the browser/server.

I Do · original lesson diagrams

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A browser sends an HTTP request using TCP/IP over Ethernet. Describe encapsulation from top to bottom.

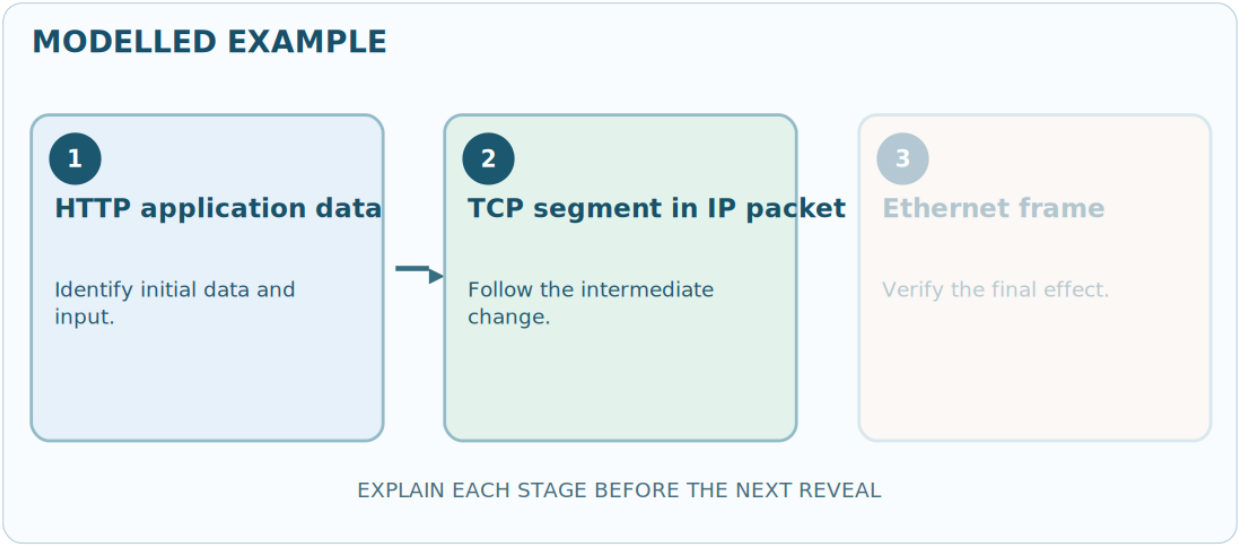
Identify the relevant information and make a prediction.



MODELLED EXAMPLE: HTTP application data, TCP segment in IP packet, Ethernet frame

A browser sends an HTTP request using TCP/IP over Ethernet. Describe encapsulation from top to bottom.

Identify the relevant information and make a prediction.



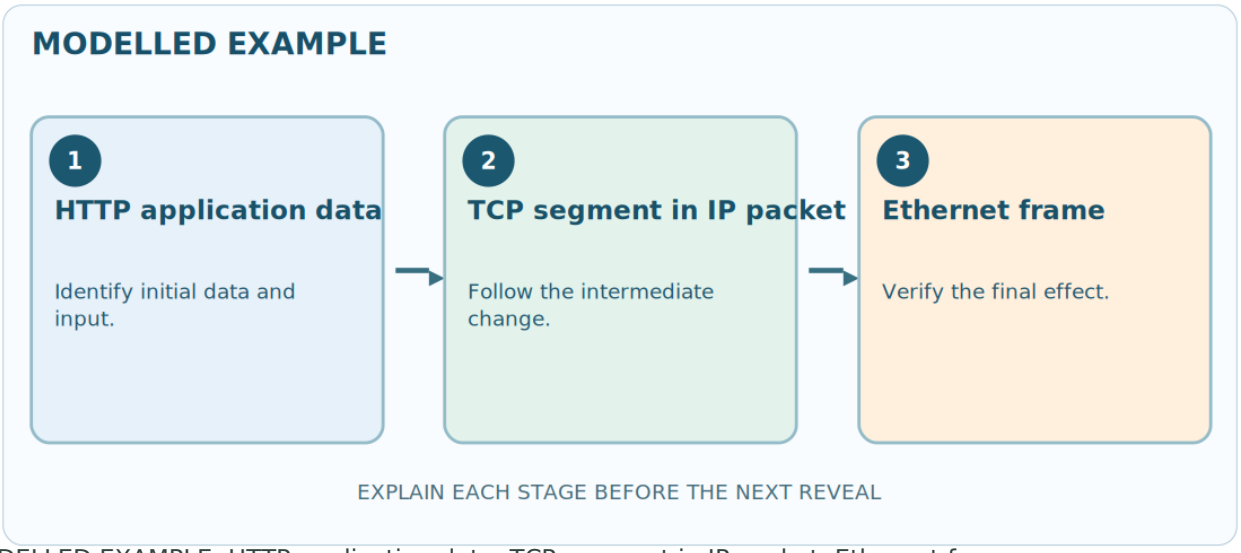
MODELLED EXAMPLE: HTTP application data, TCP segment in IP packet, Ethernet frame

I Do · original lesson diagrams

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MODELLED EXAMPLE: HTTP application data, TCP segment in IP packet, Ethernet frame

WE DO — GUIDED PRACTICE

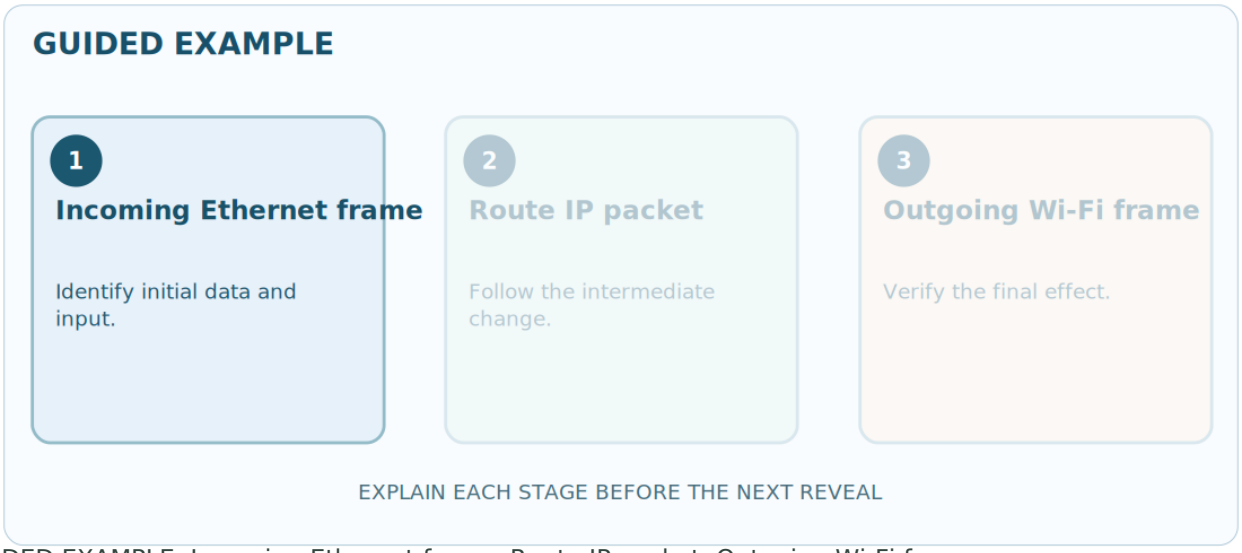
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A packet passes through a router from Ethernet into a Wi-Fi network. Which information changes?

Identify the relevant information and make a prediction.



GUIDED EXAMPLE: Incoming Ethernet frame, Route IP packet, Outgoing Wi-Fi frame

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Incoming Ethernet frame Identify the relevant information and make a prediction.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Name four layers of the TCP/IP stack and give one role for each. [8]

Question 2

Describe encapsulation of application data over TCP/IP and Ethernet. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain what a router changes when forwarding a packet to a different link technology. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Every router preserves the same link-layer frame all the way to the destination.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

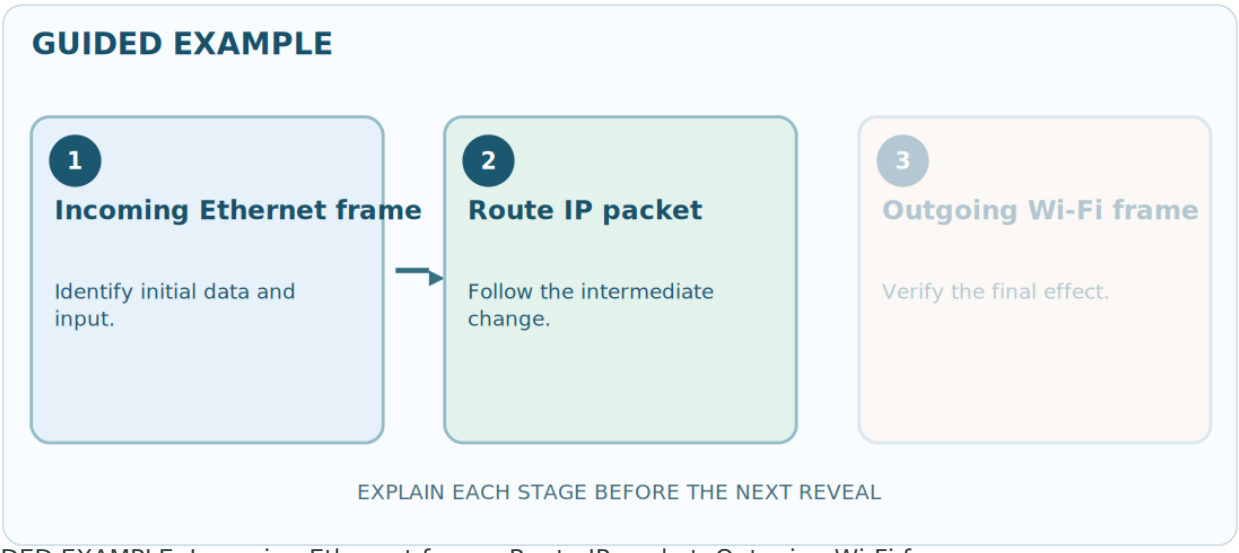
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A packet passes through a router from Ethernet into a Wi-Fi network. Which information changes?

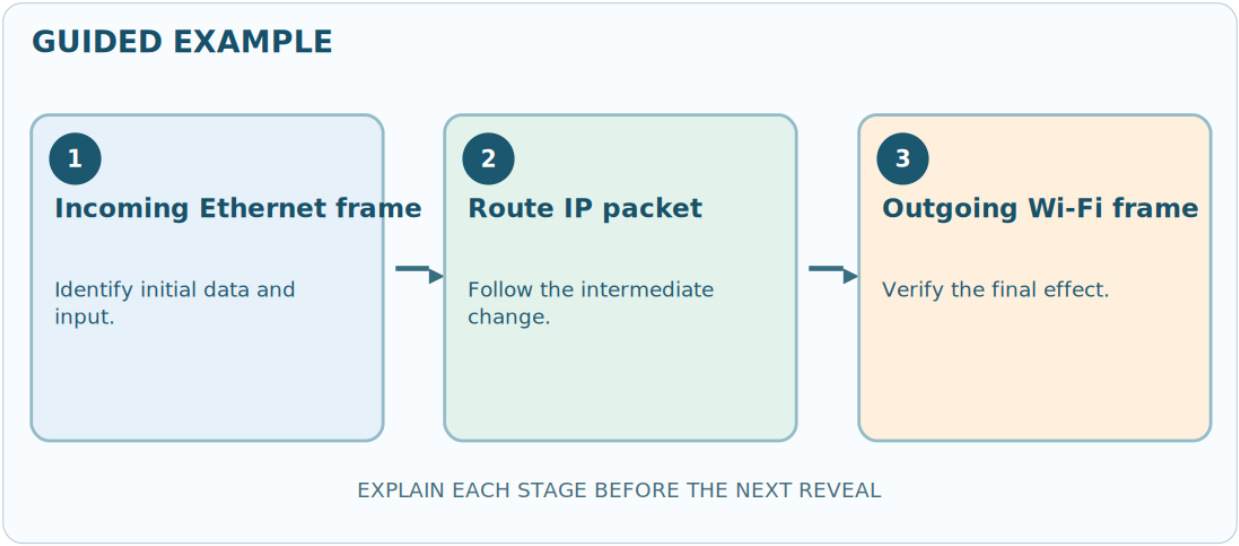
Identify the relevant information and make a prediction.



GUIDED EXAMPLE: Incoming Ethernet frame, Route IP packet, Outgoing Wi-Fi frame

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Identify the relevant information and make a prediction.



GUIDED EXAMPLE: Incoming Ethernet frame, Route IP packet, Outgoing Wi-Fi frame

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Route IP packet

Explain the change using correct technical terms.

Outgoing Wi-Fi frame

Router removes the incoming link-layer framing, processes/forwards the IP packet and builds new outgoing link-layer framing. IP hop limit/TTL normally decreases; end-to-end transport data need not change.

IMPROVE: Explain one correction to your own reasoning.